

NT L5 Lab 2026Z Final Project

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Group:L5

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Subject: Network Technologies

1. Introduction / Purpose

This project was implemented in Cisco Packet Tracer to design and configure a small enterprise network. The topology consists of four separate IPv4 LANs and two routers connected through a point-to-point WAN link. Static routing was configured to enable inter-network communication. In addition, a DNS/HTTP server was deployed to provide name resolution and web access using the domain **ronikevibeden.com**.

Network k	Network Address	Mask	Broadcast	Router Interface (Gateway)	Connected Devices (IP)
Network -I	192.168.1.0	255.255.255.0	192.168.1.25 5	192.168.1.25 4	PC0: 192.168.1.1 / PC1: 192.168.1.2 / PC2: 192.168.1.3
Network -II	192.168.2.0	255.255.255.0	192.168.2.25 5	192.168.2.25 4	PC3: 192.168.2.1 / PC4: 192.168.2.2 / PC5: 192.168.2.3

Network -III	192.168.3.0	255.255.255.0	192.168.3.255	192.168.3.254	PC6: 192.168.3.1 / PC7: 192.168.3.2 / PC8: 192.168.3.3
Network -IV	192.168.75.0	255.255.255.0	192.168.75.255	192.168.75.254	PC9: 192.168.75.1 / PC10: 192.168.75.2 / PC11: 192.168.75.3 / Server: 192.168.75.10
WAN Link	222.2.2.0	255.255.255.252	222.2.2.3	R0: 222.2.2.1 – R1: 222.2.2.2	Router0 ↔ Router1

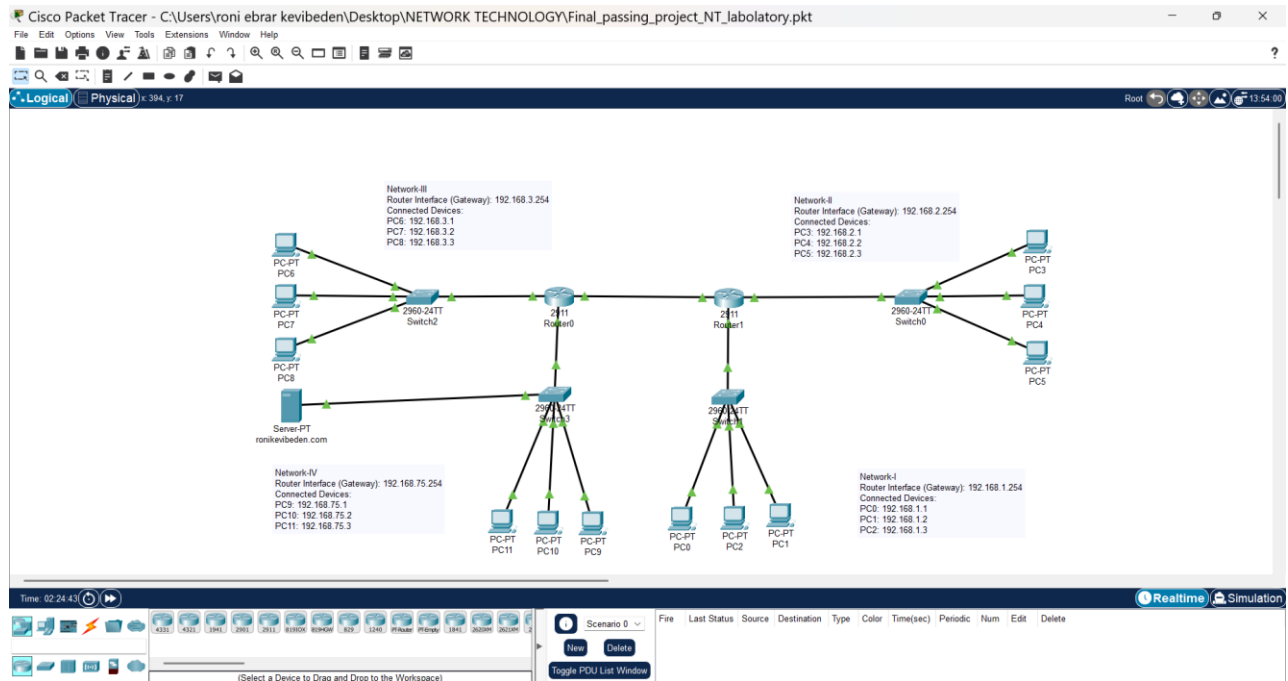
List of devices used

- **Routers:** 2 × Cisco 2911 (Router0, Router1)
- **Switches:** 4 × 2960-24TT (Switch0, Switch1, Switch2, Switch3)
- **PCs:** 12 × PC-PT (PC0–PC11)
- **Server:** 1 × Server-PT (RoniKevibeden.com)

2. Logical Topology

(Insert SS: Logical diagram)

Figure X shows the logical topology of the network. Each LAN is connected to a switch, and each switch is connected to a router interface that acts as the default gateway for that LAN. Router0 and Router1 are interconnected via the WAN network **222.2.2.0/30**.



3.DNS and Web Server Configuration

A server was configured to provide DNS and HTTP services. The DNS service includes an **A record** mapping the domain name **ronikevibeden.com** to the server IP address **192.168.75.10**.

(Insert SS: Server → Services → DNS showing A record)

Text to add under the DNS SS:

The DNS service was enabled and an A record was created for **ronikevibeden.com** → **192.168.75.10**, allowing clients to resolve the domain name to the correct IP address.

(Insert SS: Server → Services → HTTP “On”)

Text to add under the HTTP SS:

The HTTP service was enabled to allow clients to access the hosted webpage via IP address or domain name.

ronikevibeden.com

Physical Config Services Desktop Programming Attributes

Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.75.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.75.254

DNS Server: 192.168.75.10

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:42FF:FE3C:C47D

Default Gateway: /

DNS Server: /

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username: /

Password: /

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Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.2.254

DNS Server: 192.168.2.200

Start IP Address: 192.168.75.0

Subnet Mask: 255.255.255.0

Maximum Number of Users: 253

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.2...	192.168.2...	192.168.75.0	255.255.2...	253	0.0.0.0	0.0.0.0

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Physical Config Services Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Global Settings

Display Name: ronikevibeden.com

Gateway/DNS IPv4

☐ DHCP ☒ Static

Default Gateway: 192.168.75.254

DNS Server: 192.168.75.10

Gateway/DNS IPv6

☐ Automatic ☒ Static

Default Gateway: /

DNS Server: /

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Physical Config Services Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Port Status: ☒ On

Bandwidth: 100 Mbps 10 Mbps ☒ Auto

Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address: 0001.423C.C47D

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.75.10

Subnet Mask: 255.255.255.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:42FF:FE3C:C47D

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4. Router Configuration

ROUTER 0

Router0

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0  192.168.3.254  YES manual up            up
GigabitEthernet0/1  192.168.75.254  YES manual up            up
GigabitEthernet0/2  222.2.2.1       YES manual up            up
Vlan1          unassigned      YES unset  administratively down down

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.1.0/24 [1/0] via 222.2.2.2
S    192.168.2.0/24 [1/0] via 222.2.2.2
C    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
     C    192.168.3.0/24 is directly connected, GigabitEthernet0/0
     L    192.168.3.254/32 is directly connected, GigabitEthernet0/0
C    192.168.75.0/24 is variably subnetted, 2 subnets, 2 masks
     C    192.168.75.0/24 is directly connected, GigabitEthernet0/1
     L    192.168.75.254/32 is directly connected, GigabitEthernet0/1
C    222.2.2.0/24 is variably subnetted, 2 subnets, 2 masks
     C    222.2.2.0/30 is directly connected, GigabitEthernet0/2
     L    222.2.2.1/32 is directly connected, GigabitEthernet0/2

Router>
```

CopyPaste

☐ Top

Router0

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0  192.168.3.254  YES manual up            up
GigabitEthernet0/1  192.168.75.254  YES manual up            up
GigabitEthernet0/2  222.2.2.1       YES manual up            up
Vlan1          unassigned      YES unset  administratively down down

Router#

Router con0 is now available.

Press RETURN to get started.

Router>show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0  192.168.3.254  YES manual up            up
GigabitEthernet0/1  192.168.75.254  YES manual up            up
GigabitEthernet0/2  222.2.2.1       YES manual up            up
Vlan1          unassigned      YES unset  administratively down down

Router>
```

CopyPaste

☐ Top

Router0 – Static routing

- ip route 192.168.1.0 255.255.255.0 222.2.2.2
- ip route 192.168.2.0 255.255.255.0 222.2.2.2

ROUTER1

The image displays two screenshots of the Router1 CLI interface, showing the configuration and status of the router.

Top Screenshot: The CLI shows the output of the `show ip interface brief` and `show ip route` commands.

Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	222.2.2.2	YES	manual	up	up
GigabitEthernet0/1	192.168.1.254	YES	manual	up	up
GigabitEthernet0/2	192.168.2.254	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Router>show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/1
L    192.168.1.254/32 is directly connected, GigabitEthernet0/1
L    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/2
L    192.168.2.254/32 is directly connected, GigabitEthernet0/2
S    192.168.3.0/24 [1/0] via 222.2.2.1
S    192.168.75.0/24 [1/0] via 222.2.2.1
S    222.2.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    222.2.2.0/30 is directly connected, GigabitEthernet0/0
L    222.2.2.2/32 is directly connected, GigabitEthernet0/0
```

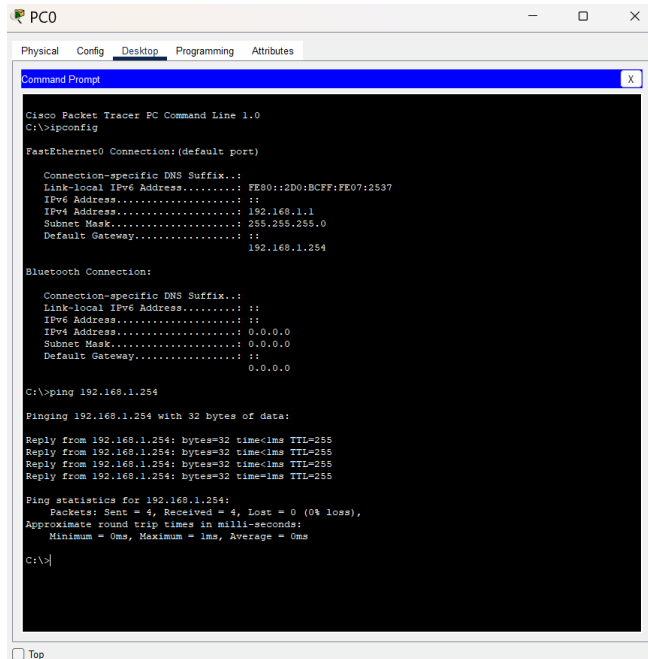
Bottom Screenshot: The CLI shows the output of the `show ip interface brief` command, indicating that the router is now available.

Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	222.2.2.2	YES	manual	up	up
GigabitEthernet0/1	192.168.1.254	YES	manual	up	up
GigabitEthernet0/2	192.168.2.254	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

5. Test of the completed network. Confirmation of the results obtained

PC0 (Connected to Switch 1):



The screenshot shows the Cisco Packet Tracer PC Command Line interface for PC0. The 'Desktop' tab is active, displaying the 'Command Prompt' window. The prompt shows the configuration of the FastEthernet0 interface with an IPv6 address of 192.168.1.1 and a subnet mask of 255.255.255.0. The default gateway is set to 192.168.1.254. A ping command is executed to 192.168.1.254, resulting in four successful replies with a TTL of 255 and a round trip time of 0ms. The ping statistics show 4 packets sent, 4 received, and 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:BCFF:FE07:2537
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.1.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
    0.0.0.0

C:\>ping 192.168.1.254

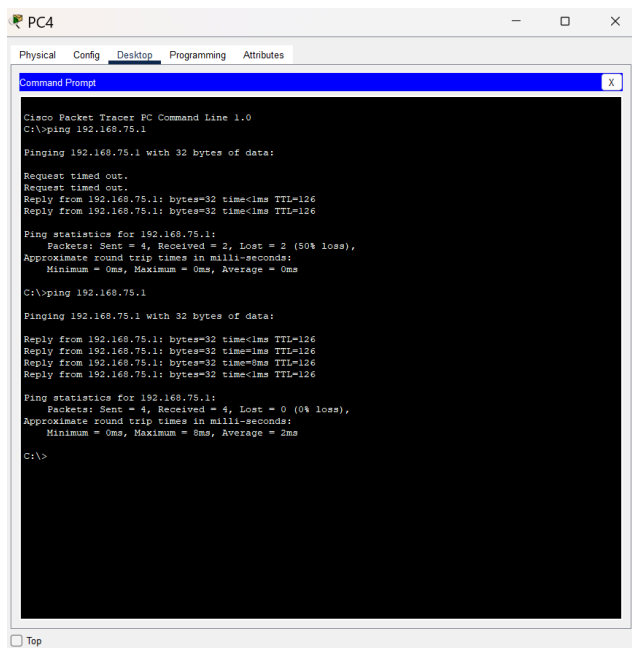
Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PC4 (Connected to Switch 0):



The screenshot shows the Cisco Packet Tracer PC Command Line interface for PC4. The 'Desktop' tab is active, displaying the 'Command Prompt' window. The prompt shows the configuration of the FastEthernet0 interface with an IPv6 address of 192.168.75.1 and a subnet mask of 255.255.255.0. The default gateway is set to 192.168.75.1. A ping command is executed to 192.168.75.1, resulting in four successful replies with a TTL of 126 and a round trip time of 0ms. The ping statistics show 4 packets sent, 4 received, and 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.75.1

Pinging 192.168.75.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.75.1: bytes=32 time<1ms TTL=126
Reply from 192.168.75.1: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.75.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.75.1

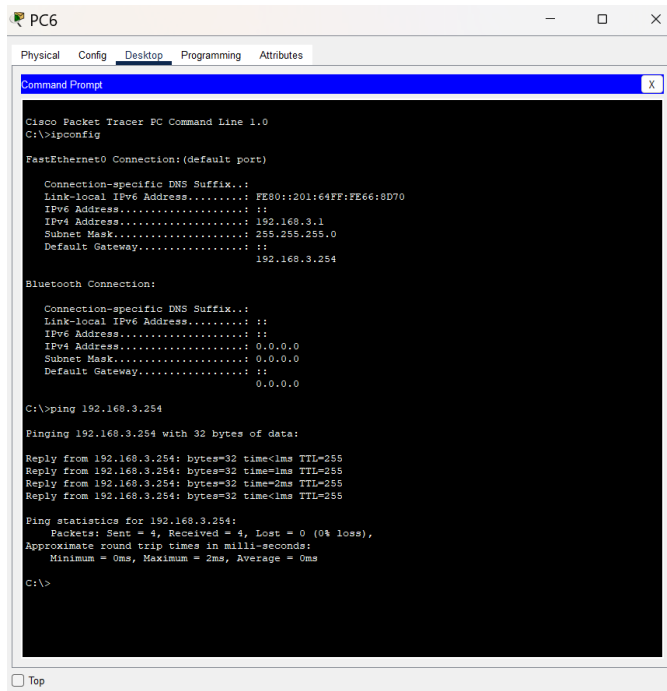
Pinging 192.168.75.1 with 32 bytes of data:

Reply from 192.168.75.1: bytes=32 time<1ms TTL=126
Reply from 192.168.75.1: bytes=32 time<1ms TTL=126
Reply from 192.168.75.1: bytes=32 time=0ms TTL=126
Reply from 192.168.75.1: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.75.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 2ms

C:\>
```

PC6 (Connected to Switch 2):



The screenshot shows the Command Prompt window for PC6 in Cisco Packet Tracer. The window title is 'PC6' and it has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt shows the output of the 'ipconfig' command, displaying network configuration for the FastEthernet0 interface. It includes IPv4 and IPv6 addresses, subnet mask, and default gateway. Below the configuration, the output of the 'ping 192.168.3.254' command is shown, indicating successful connectivity with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:64FF:FE66:8D70
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.3.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.3.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 192.168.3.254

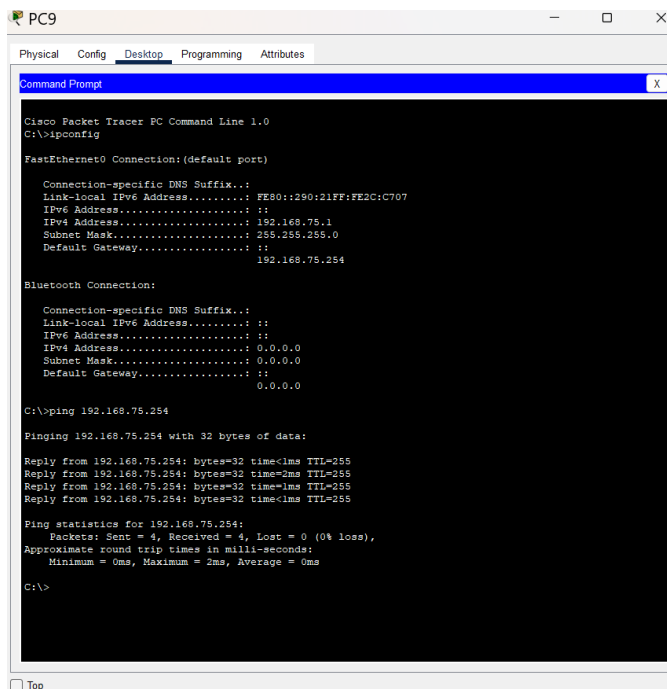
Pinging 192.168.3.254 with 32 bytes of data:

Reply from 192.168.3.254: bytes=32 time<1ms TTL=255
Reply from 192.168.3.254: bytes=32 time=2ms TTL=255
Reply from 192.168.3.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.3.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

PC9 (Connected to Switch 3):



The screenshot shows the Command Prompt window for PC9 in Cisco Packet Tracer. The window title is 'PC9' and it has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt shows the output of the 'ipconfig' command, displaying network configuration for the FastEthernet0 interface. It includes IPv4 and IPv6 addresses, subnet mask, and default gateway. Below the configuration, the output of the 'ping 192.168.75.254' command is shown, indicating successful connectivity with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::290:21FF:FE2C:C707
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.75.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.75.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

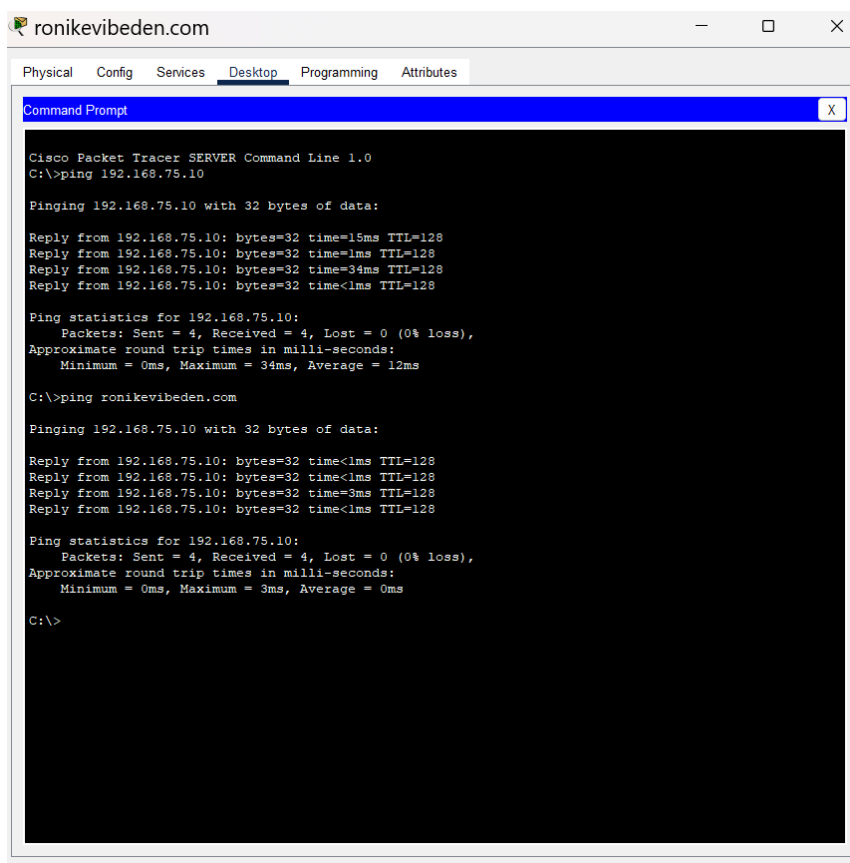
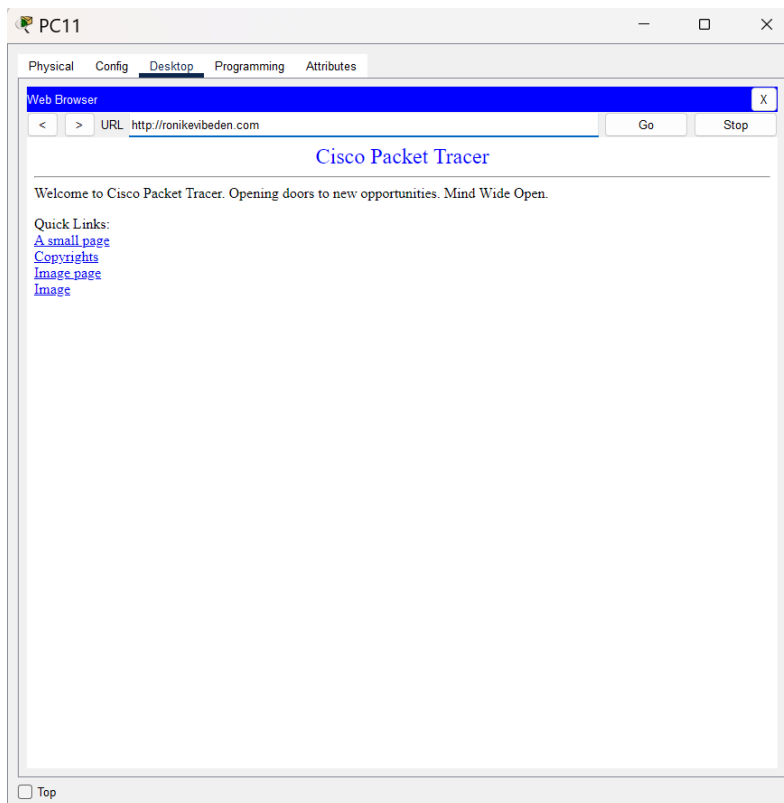
C:\>ping 192.168.75.254

Pinging 192.168.75.254 with 32 bytes of data:

Reply from 192.168.75.254: bytes=32 time<1ms TTL=255
Reply from 192.168.75.254: bytes=32 time=2ms TTL=255
Reply from 192.168.75.254: bytes=32 time=1ms TTL=255
Reply from 192.168.75.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.75.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

6.Conclusion

This project successfully implemented a multi-LAN network in Cisco Packet Tracer using two routers and static routing. All LANs were able to communicate with each other through the WAN link, as verified by ping tests and routing table outputs. DNS and HTTP services were configured on the server, enabling domain name resolution and web access via **ronikevibeden.com**. The collected screenshots demonstrate correct configuration and successful end-to-end operation.